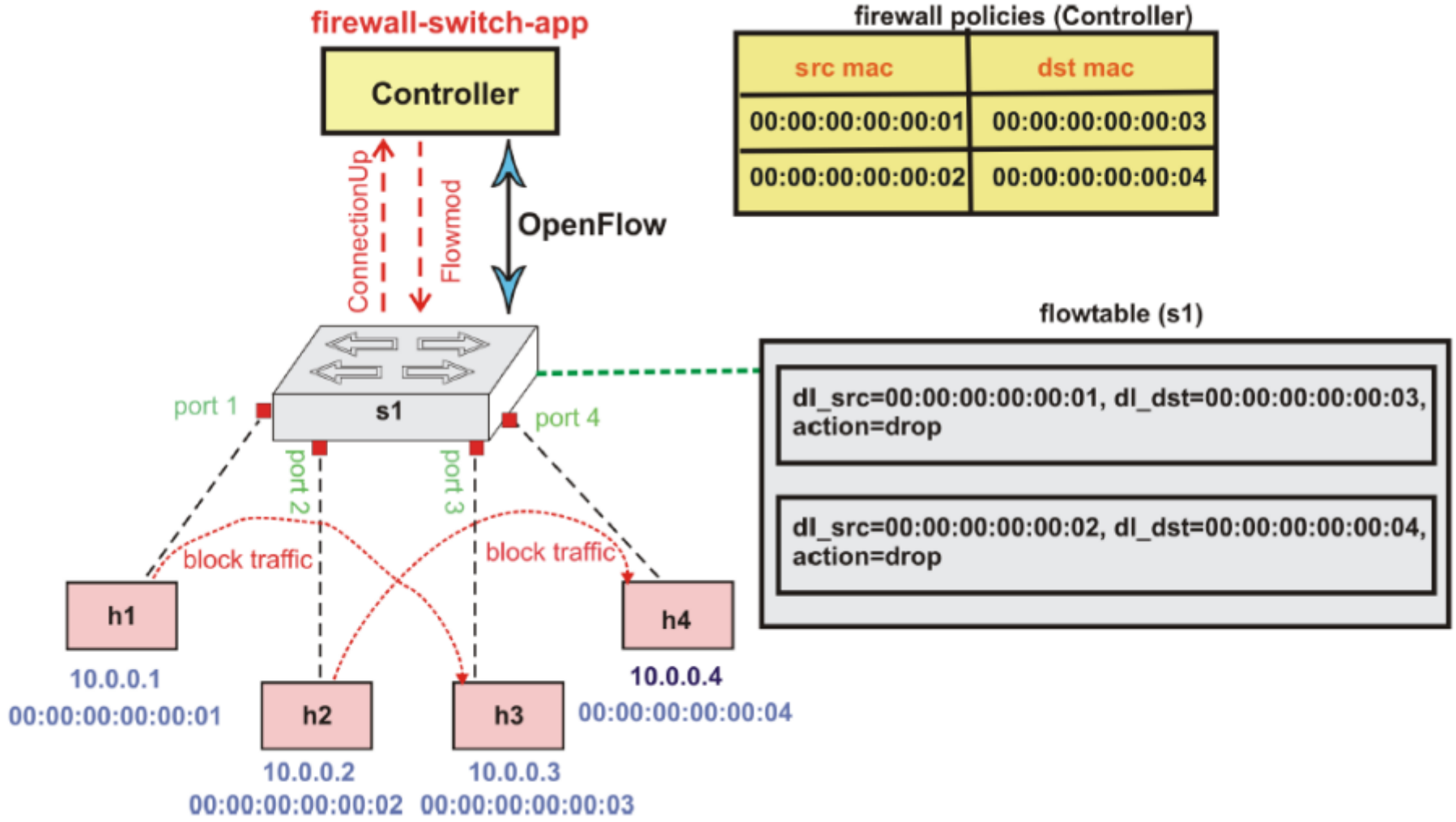

firewall-switch-app
Controller
ConnectionUp
Flowmod
OpenFlow
s1
port 1
port 2
port 3
port 4
block traffic
block traffic
h1
h2
h3
h4
10.0.0.1
00:00:00:00:00:01
10.0.0.2
00:00:00:00:00:02
10.0.0.3
00:00:00:00:00:03
10.0.0.4
00:00:00:00:00:04
firewall policies (Controller)
src mac
dst mac
00:00:00:00:00:01
00:00:00:00:00:03
00:00:00:00:00:02
00:00:00:00:00:04
flowtable (s1)
dl_src=00:00:00:00:00:01, dl_dst=00:00:00:00:00:03, action=drop
dl_src=00:00:00:00:00:02, dl_dst=00:00:00:00:00:04, action=drop

- **Assignment: Implement an SDN-Based Layer-2 Firewall using POX, OpenFlow, and Mininet:**
- Consider a Mininet network consisting of four hosts (h_1, h_2, h_3, h_4) connected to a single OpenFlow switch s_1.
- The switch is controlled by a POX SDN controller. Your task is to implement a **Layer-2 firewall application** in the controller.
- The firewall policy is specified externally in a policy file.
- Each rule contains a **source MAC address and destination MAC address**.

- If an Ethernet frame matches one of these prohibited sourc-destination MAC pairs, the switch must **drop the frame**.
- Traffic that does not match a firewall rule must be forwarded normally using learning-switch behaviour.
- When the OpenFlow switch connects to the controller, the controller shall read all firewall policies, convert them into OpenFlow flow entries, and install the corresponding **drop rules proactively** in the switch.
- The controller must subsequently learn host MAC-to-port mappings and forward allowed traffic efficiently.

- Part 2: **Intent-Based Networking + Generative AI + SDN** :
- We know that AI can operate in the management/control layer, while deterministic networking mechanisms remain in the data plane.

- Give an LLM (Phi3) the database of MAC:
- h1 = 00:00:00:00:00:01
- h2 = 00:00:00:00:00:02
- h3 = 00:00:00:00:00:03
- h4 = 00:00:00:00:00:04

- Then, provide the administrator's request: (it’s a natural language request):
- Block h1 from sending traffic to h3.
- Block h2 from sending traffic to h4.
- Allow everything else.
- **LLM output: JSON (Defined rule, src, dst address , action)**
- Then, the POX controller converts to
- src MAC = 00:...:01
- dst MAC = 00:...:03
- action = DROP into PYTHON based RULE.

- Block h1 from h3→ LLM (Phi3)→h1-MAC → h3-MAC → DROP→ POX Controller→OpenFlow→ Switch

- Natural-language intent → Phi-3 → JSON/CSV policy → existing POX firewall.